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Characteristics of Ni-Cu-PGE Mineralization and Genesis of the Jinchuan Deposit, Northwest China

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Abstract

The Jinchuan Ni-Cu sulfide deposit is hosted by a northwest-striking ultramafic body located at the southwest margin of the Sino-Korean platform, China. Sulfide mineralization occurs in the lower part of the intrusion and comprises three main orebodies, 24 and 1 in the west, and 2 in the east of the intrusion. Three types of sulfide ores are identified: net textured, disseminated, and a small amount of massive ore which occurs only in orebody 2. The principal ore minerals are pyrrhotite, pentlandite, violarite, chalcopyrite, and cubanite.

A systematic study has been made of the S, Ni, Cu, Co, platinum-group elements (PGE), and Au of the ore samples. A lateral zonation is defined by these elements in the two western orebodies. The concentrations of Ni, Cu, and PGE increase from the margins toward the center of the orebodies. This trend is also present when the metal elements are normalized to the S content of the samples. A vertical variation of elements is present in the east orebody where the amounts of Ni, Cu, and PGE, as well as S, decrease upward. However, when normalized to the S content, PGE increase upward, but Ni and Cu remain more or less constant. A plot of PGE versus S clearly shows three sample sets: (1) samples from orebody 24 have lower S contents, mostly less than 5 wt percent, and their PGE contents roughly increase with that of sulfur; (2) orebody 1 has higher PGE contents, but the PGE plots are more scattered, suggesting multiple processes during PGE mineralization; and (3) orebody 2 has very low PGE concentrations.

Ore samples studied show an average Ni/Cu ratio of 2, and (Pt + Pd)/(Ru + Ir + Os) ratios of 5 to 30, with most values falling in the range of 10 to 15. These are typical features of sulfide ores that relate to mafic magma. There is little difference in the (Pt + Pd)/Os + Ir + Ru) ratios and the chondrite-normalized PGE distribution patterns between the PGE-rich and PGE-poor ores, suggesting that they originated from magmas with similar major element compositions.

The close association of PGE enrichment with sulfide mineralization and the strong correlations exhibited in all ore types between S and Ni, Co, and PGE indicate that the ores are the consequence of the segregation of sulfide from silicate magmas. The high PGE contents of ore-bodies 24 and 1 and the low PGE contents of orebody 2 suggest that two batches of sulfide liquid were involved. The PGE-rich sulfide liquid equilibrated with a relatively primitive silicate magma whereas the PGE-poor sulfide liquid equilibrated with a silicate magma from which PGE had been depleted by prior (probably fractional) sulfide segregation.

The low values of Ni/Cu ratios (about 2) and high values of (Pt + Pd)/(Ru + Ir + Os) ratios (10–15) of the samples, together with the chemical composition and rare earth element data of the intrusive rocks provide convincing evidence that the initial magma of the intrusion was gabbroic rather than ultramafic.

Introduction

THE Jinchuan sulfide deposit is a major source of Ni and Cu. It contains more than 500 million metric tons of ore grading 1.2 wt percent Ni and 0.7 wt percent Cu. This ore reserve has made Jinchuan the second largest Ni deposit in the world, after that of Sudbury, Canada (Ross and Travis, 1981). Platinum-group elements (PGE) and Au are also very important economic components of the Jinchuan Ni-Cu deposit, which is the only active PGE producer in China. To date, numerous PGE-enriched lenses, which are defined as areas where Pt or Pd exceeds 1 ppm, have been defined within the sulfide orebodies and range up to 500 m long and 50 m thick. Although the PGE and Au grades of the sulfide ores are not particularly high (with an average PGE + Au grade of about 1 ppm), the total PGE reserves in the deposit are considerable due to its huge size.

The geology and Ni-Cu sulfide mineralization of the Jinchuan deposit have been documented, and the deposit is considered to be ultramafic related, based on the characteristics of the host rocks (S.G.U.,[1] 1984). Assays for Pt and Pd were obtained during

[1] Sixth Geological Unit of the Gansu Geological Survey, China.

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exploration and mining, but virtually no geochemical studies were carried out on PGE.

In this study of the Jinchuan ores, we investigate the characteristics of Ni-Cu sulfide mineralization, the distribution of PGE, the relationship between the PGE and Ni sulfide mineralizations, and the role of magmatic sulfide and hydrothermal fluids in concentrating PGE. We also present a genetic interpretation for the PGE mineralization process at the Jinchuan deposit.

Geology of the Deposit

The Jinchuan Ni-Cu deposit is located in an uplifted terrane at the southwest margin of the Sino-Korean platform (Fig. 1). The uplifted terrane strikes northwest and comprises mainly early Proterozoic migmatites, biotite gneisses, and marbles (S.G.U., 1984). It is bounded by two regional faults on both sides. A series of Precambrian mafic-ultramafic intrusions is distributed in this terrane along the two

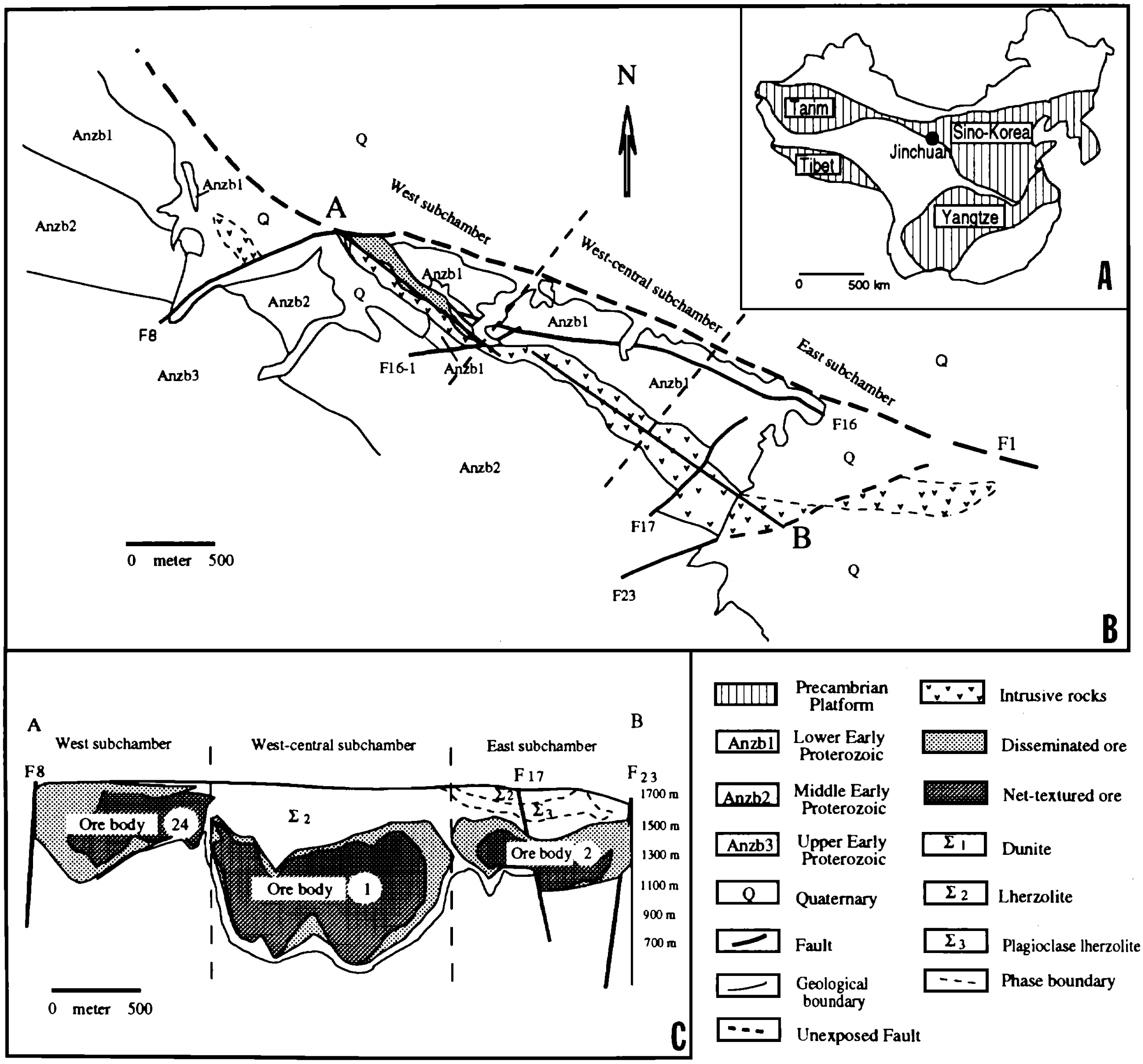


FIG. 1. Geology of the Jinchuan deposit. A. Location of Jinchuan in China. B. Geologic map of the Jinchuan intrusion and the Ni-Cu deposit (modified from Jia, 1986). C. Longitudinal sections of the Jinchuan deposit.

faults. The Jinchuan intrusion, which hosts the Ni-Cu deposit, intrudes at the northern margin of the uplifted terrane. It is a northwest-elongated ultramafic body that is 6,000 m long and 300 m wide (Jia, 1986; Chai and Naldrett, 1992). The general geology and structure of the Jinchuan intrusion were first described in detail by S.G.U. (1984) and the Ni-Cu mineralization by Jia (1986). Recently, Chai and Naldrett (1992) carried out systematic petrological and geochemical studies on the Jinchuan intrusive rocks. They divided the igneous body into three magma subchambers, the west, west-central, and east (Fig. 1B), and their geochemical study led to the conclusion that the Jinchuan intrusion represents only the ultramafic cumulates of a mafic magma. Because of the lack of published information and difficulty in access to the Chinese literature, we present a detailed description of the geology of the Jinchuan deposit.

The Ni-Cu ore of the Jinchuan deposit is defined as rocks that contain greater than 0.5 wt percent Ni (S.G.U., 1984). Numerous ore lenses have been delineated in the intrusion, mainly in the lower part of the igneous body. Most of them are very small in size except for three, orebodies 24, 1, and 2, which correspond to the three magma subchambers and account for more than 90 percent of the Ni-Cu-PGE reserves of the deposit (Fig. 1C).

The west subchamber extends from the west end of the intrusion to the intersection with the northeast-striking fault $F_{16\text{-}1}$ (Fig. 1B). It comprises two sections: the western and the eastern.

The western section occurs on the western side of fault F_8 and is offset 900 m southwest from the main intrusive body. It is a wedge-shaped block about 600 m long, widening southeast to about 250 m at F_8; the depth of the base increases from 200 to about 600 m to the east. Three rock types, dunite, lherzolite, and olivine pyroxenite, occur in this portion (Fig. 2). Dunite, which is commonly less than 50 m thick, forms the core of the intrusive body and is surrounded by lherzolite which is present on both flanks of the dunite and also in the upper part of the intrusion. Olivine pyroxenite occurs as thin zones up to 30 m on the margins of the igneous body. Almost all the intrusive rocks except olivine pyroxenite in this block host sulfide mineralization, but only a small portion of the ore contains greater than 1 wt percent Ni.

The eastern section of the west subchamber extends from F_8 eastward to $F_{16\text{-}1}$. It is about 1,500 m long and narrows from 300 m wide at the western end to less than 30 m at the eastern end (Fig. 2). The dunite core is up to 100 m thick at depth and narrows upward. Lherzolite occurs in the upper part of the intrusion and envelops dunite at depth. Olivine pyroxenite occurs discontinuously and forms a thin marginal zone. Locally, plagioclase lherzolite crops out on the surface as thin bands, which commonly pinch out in less than 100 m.

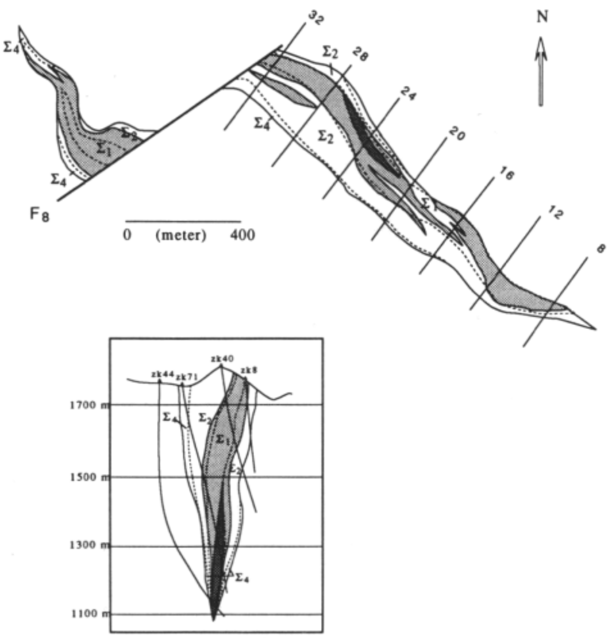


FIG. 2. Geologic plan of the 1,250-m level and cross section of the drill grid 24 of orebody 24. The drill grid is selectively shown as parallel lines in the upper diagram; the grid lines are at 50-m intervals and are numbered continuously; zk plus number indicates the drill holes. See Figure 1 for legend.

Orebody 24 is the major orebody in the west subchamber. It is a tabular-shaped mass subparallel to the wall of the intrusion. The orebody is about the length of the west subchamber with a width of about 100 to 200 m (Fig. 2). Sulfide ore is exposed on the surface near F_8 at the western end of the eastern block, which is the site of the open pit of the mine, and extends eastward for about 600 m where it plunges underground. The sulfide ore in the open pit is usually low grade with an Ni content less than 1 wt percent. At the contact between the orebody and F_8, sulfide ores are strongly oxidized, and the oxidation zone extends to about a 120-m depth. The silicates and sulfides are replaced by talc, magnesite, violarite, and hematite.

The west-central subchamber extends from $F_{16\text{-}1}$ in the west about 1,700 m to the east; the width of the intrusive body does not vary visably on the surface, but the base of the intrusion rises from a depth of more than 1,000 to about 500 m (Fig. 1C). The rock types and their distribution are similar to those in the west subchamber except that the sulfide-bearing dunite in the west-central subchamber is much thicker than that in the west subchamber.

The west-central subchamber hosts the largest orebody (1) of the deposit which accounts for more than 50 percent of the total reserves of both Ni and Cu. The orebody is tabular in shape and constitutes almost the entire amount of igneous rocks at depth (Fig. 3). It occurs from a depth of about 200 to more than 1,100 m, for a strike length of 1,500 m, and a width of about 150 m. Most of the sulfide ores in this orebody are high grade (1–4 wt % Ni) and are quite evenly distributed. Because division of sulfide ore is defined by its Ni grade, the transition between the ore and igneous rocks is gradational.

The east subchamber comprises all the portions east of the west-central subchamber (Fig. 1B). The intrusion becomes wider and shallower in this part, and the cross section becomes trumpet shaped (Fig. 4). Two major northeast-striking faults (F_{17} and F_{23}) cut the subchamber into three sections. F_{17} displaces the central block vertically downward 90 to 150 m and laterally southwest 150 to 260 m. This has brought the wider upper part of the central block to

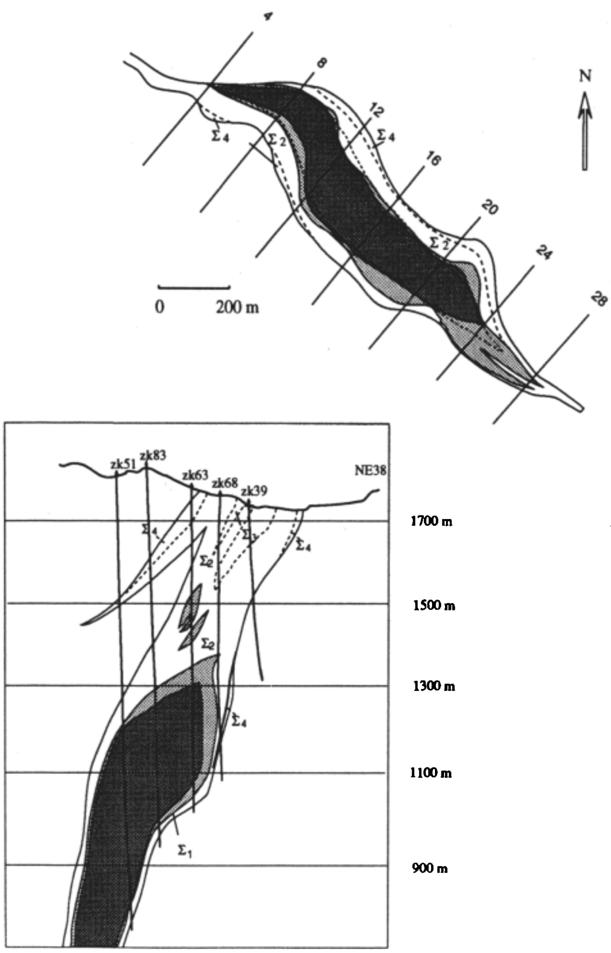


FIG. 3. Geologic plan of the 1,250-m level and cross section of the drill grid 14 of orebody 1. The deeper part of the orebody is under exploration. See Figure 1 for legend.

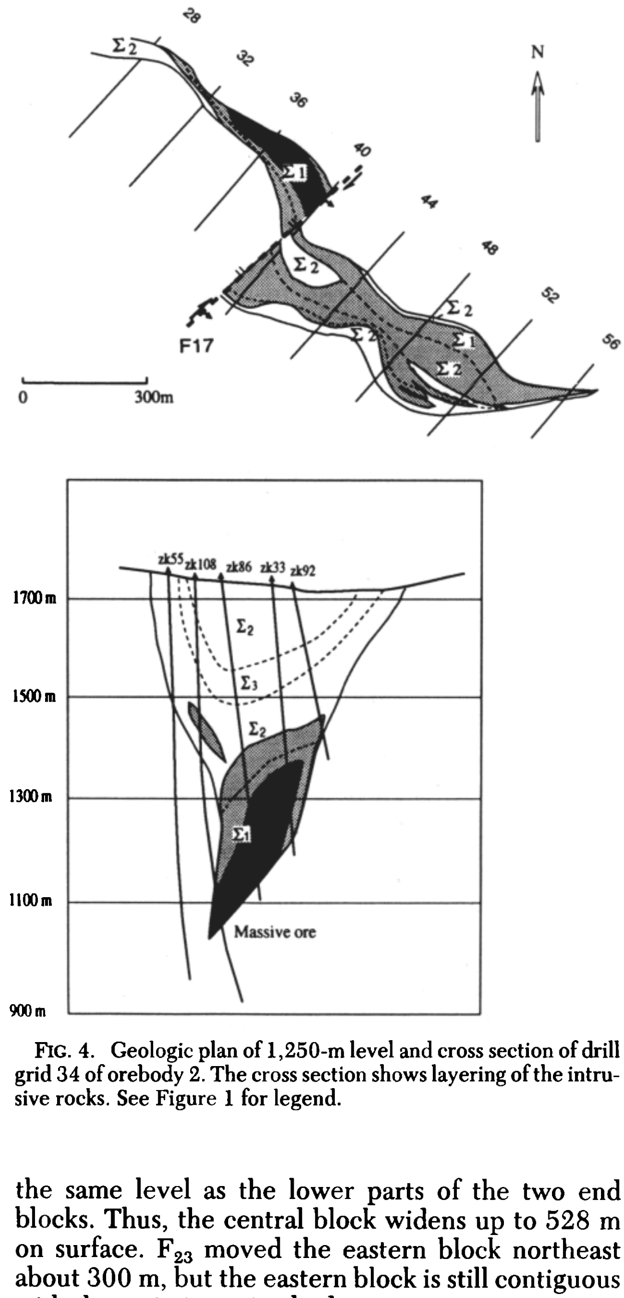


FIG. 4. Geologic plan of 1,250-m level and cross section of drill grid 34 of orebody 2. The cross section shows layering of the intrusive rocks. See Figure 1 for legend.

the same level as the lower parts of the two end blocks. Thus, the central block widens up to 528 m on surface. F_{23} moved the eastern block northeast about 300 m, but the eastern block is still contiguous with the main intrusive body.

Orebody 2 is located at the base of this subchamber. It is 1,300 m long with a width of 118 m (Fig. 4). Generally, the orebody defines the base of the intrusion and dips slightly to the southwest at 25° to 60°. The grade of the sulfide ore decreases toward the east and is usually less than 1 wt percent in the portion east of F_{23}.

Ore Petrology

Several ore types are present in the deposit. These include mainly net-textured, disseminated, and massive ores in a decreasing order. Locally, metasomatic ore is observed at the contact of marbles with the igneous rocks.

Net-textured ore, in which sulfide fills all the interstitial voids of the olivine cumulates, these connecting them and forming a continuous network (Fig. 5C), is the dominant ore type of the deposit. The Ni grade usually ranges from 1 to 4 wt percent, with most at about 2 wt percent. Net-textured ore commonly forms ore lenses of various sizes in the core or lower part of the sulfide orebodies. This ore type accounts for less than 10 percent of the nickel reserve in orebody 24. There, it occurs 50 m below surface in the eastern portion of the west subchamber and is usually less than 40 m thick. The ore lens is close to the footwall side of the orebody. However, net-textured ore dominates orebody 1. Except for the top portion and the thin marginal zones, the rest is all net-textured ore. This makes the net-textured ore lens more than 1,500 m long and up to 150 m thick. The net-textured ore lens in both orebodies 24 and 1 is enveloped by disseminated ore and the two types of ores are gradational to each other. Net-textured ore occurs at the base of orebody 2. It is locally cut by massive sulfide veins near the base of the ore lens (Fig. 4).

Disseminated ore is the part of the orebody where interstitial sulfide does not form a continuous network; in most cases the sulfide ore occurs as discrete patches in the silicate rocks (Fig. 5B). This ore type usually contains less than 1 wt percent Ni. It occurs either as small ore lenses in the upper part of the igneous body or at both ends and on the flanks of the net-textured ore lens, forming an envelope around it. In the western portion of orebody 24 and the eastern portion of orebody 2, disseminated ore is the only ore type. However, in terms of Ni and PGE reserves, this ore type only accounts for a small portion of the reserves.

Massive ore (Fig. 5A), which contains 4 to 9 wt percent Ni, is present only in orebody 2. It commonly occurs as small aggregates or irregular veins with the same strike as the main orebody. These aggregates are commonly located at the base of the orebody, grading into net-textured ore and disseminated ore upward or toward the margin. In this case, the boundary between the massive and the other ore types is gradational. Massive ore veins commonly occur in the net-textured ore and have a width of about 1 to 5 m. Some small massive ore veins (1 m) cut other sulfide ores and extend into the country rocks, although they do not extend very far (a few meters). Xenoliths of intrusive rocks, gneisses, marbles, and other types of sulfide ores commonly occur in the massive sulfides. They are most abundant near the base of the orebody where they are called breccia ore if the xenoliths account for more than 10 percent of the volume. The presence of breccia ore may suggest that massive sulfide was emplaced dynamically.

Metasomatic ore is locally present at the contact between the intrusion and the marble, and also around xenoliths of the marble. This type of ore occurs mainly in the eastern part of orebody 24 near the margin of the orebody as very irregularly shaped bodies. Sulfide can occur as irregular veins or aggregates in the carbonate groundmass. Metasomatic orebodies are difficult to define in the intrusion because they

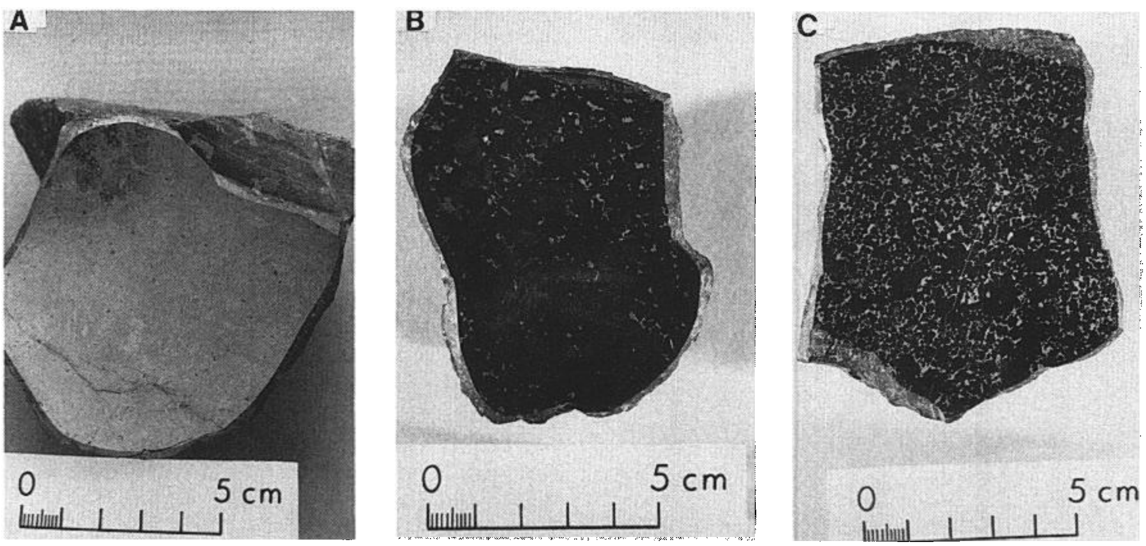


FIG. 5. Photographs showing the characteristics of the Jinchuan ore types. A. Massive ore. B. Disseminated ore; small pots of sulfides are discretely distributed within silicates. Magmatic sulfide (white) is interstitial to cumulus olivine (dark gray). C. Net-textured ore.

are scattered in marble xenoliths and some contact zones. Since it accounts for less than 2 percent of the reserves, this type of ore is not emphasized in this study.

PGE enrichment (PGE > 1 ppm) is commonly present in the net-textured ore. The average PGE concentration in the net-textured ore is about 1 ppm; that is, about two or three times that of the disseminated ore (S.G.U., 1984). In orebody 24, 13 PGE-enriched lenses have been defined, 12 of which are distributed in the net-textured ore. These PGE-enriched lenses are all parallel to the Ni-Cu sulfide orebody, with occasional irregularities along fracture zones. The fracture zones in the orebody can be extremely enriched in PGE. One fracture zone that occurs in orebody 24 contains average Pt and Pd contents of 6.15 and 1.83 ppm, respectively. The fracture zone is 1 to 1.4 m wide and extends along the strike of the Ni-Cu orebody for more than 100 m; the silicates of the fracture zone are extensively altered to serpentine, talc, and magnesite. Orebody 1 also hosts many PGE- and Au-enriched lenses. The largest one occurs in the lower part of the Ni-Cu orebody and extends along its strike for a length of about 500 m, with a maximum width of 47 m. It has an average Pt grade of 2.4 ppm, but it can be up to more than 9 ppm. The other PGE-enriched lenses are much smaller in size and lower in grade.

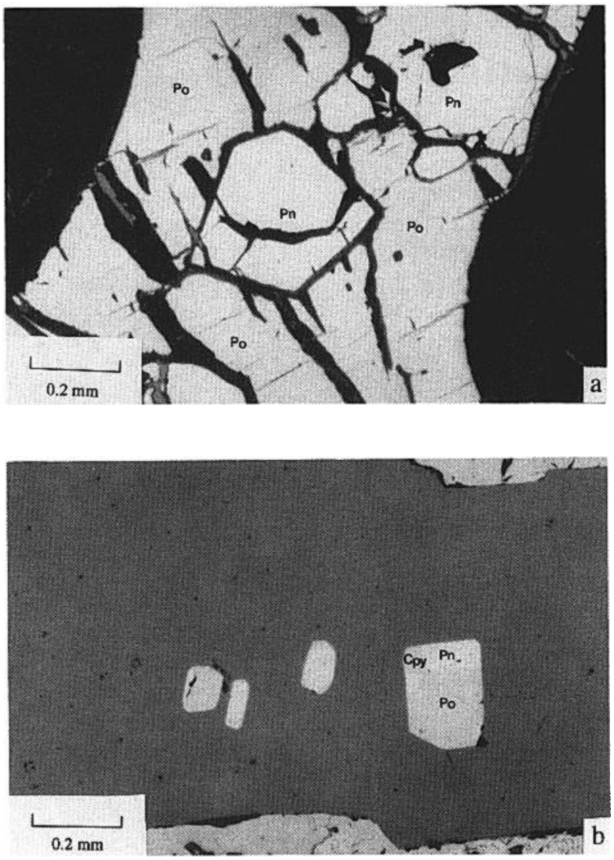


FIG. 6. Photomicrographs of sulfide minerals. A. Euhedral pentlandite (Pn) crystals are included in pyrrhotite (Po) matrix. The sulfide minerals are interstitial to cumulus olivine. B. Sulfide inclusions in cumulus chromite (Cpy: chalcopyrite). The inclusions all show rectangular shape.

Sulfide Mineralogy

Ore minerals include pyrrhotite, pentlandite (violarite), chalcopyrite, cubanite, mackinawite, and pyrite. Pyrrhotite is the dominant sulfide mineral in all the sulfide ores. It occurs mainly as anhedral or subhedral crystals with grain sizes from 0.1 to 3 mm. Pentlandite is closely associated with pyrrhotite and commonly forms subhedral to euhedral crystals (0.1–2 mm) enclosed by anhedral pyrrhotite (Fig. 6A). Rarely, exsolution flames of pentlandite occur at boundaries or fractures of pyrrhotite crystals. This mode of occurrence seems quite different from other Ni sulfide deposits such as Sudbury, Canada (Naldrett, 1984), Noril'sk, USSR (Genkin et al., 1973), and Montcalm, Canada (Barrie and Naldrett, 1989), in which flame pentlandite is very common. In massive ore, pyrrhotite and pentlandite often form a banded texture with disseminated chalcopyrite and magnetite. In this case, pentlandite is elongated along the banding, and many fractures, which are mostly filled with magnetite veins, appear perpendicular to the elongation of the crystal. This may suggest that massive sulfide was formed by stress and that the sulfide melt was flowing during crystallization. In the oxidized ore, pentlandite is replaced by violarite that occurs as very fine grains along grain margins or cleavages, forming a regular framework in the pentlandite crystals. The replacement of pentlandite by violarite increases in intensity upward in the deposit. In the open pit where sulfide ores have been intensely oxidized, almost half of the pentlandite has been replaced by violarite. Chalcopyrite occurs predominantly as anhedral assemblages disseminated within other sulfides or as very fine veinlets which crosscut olivine or pyroxene grains. Chalcopyrite is also seen to cut pyrrhotite and pentlandite. In some massive ore, fragments of pyrrhotite and pentlandite assemblages are included within a chalcopyrite-dominant groundmass, suggesting that chalcopyrite crystallized later or was introduced afterward into the massive ore. In most net-textured ore, sulfides comprise pyrrhotite, pentlandite, and chalcopyrite in decreasing order. The close association of the above sulfide minerals is best demonstrated in inclusions in olivine and chromite. It is not uncommon to see small (<0.05-mm) sulfide droplets in cumulus olivine and chromite crystals. One chromite crystal was observed to have contained four sulfide inclusions which showed perfect rectangular shape, reflecting negative morphology of the chromite (Fig. 6B). All these sulfide inclusions comprise pyrrhotite, pentlandite, and chalcopyrite in the rough proportion of 7:2:1. The existence of these sulfide inclusions clearly indicates that sulfide segregation occurred during olivine and chromite crystallization and that an important amount of copper came from the primary sulfide liquid, reflecting a high copper concentration in the original magma. Cubanite usually occurs as small bands intercalated with chalcopyrite bands perpendicular to the boundaries or fractures of the chalcopyrite. In places, it replaces pentlandite along the cleavages or fractures of the latter. Mackinawite occurs predominantly in pyrrhotite and pentlandite as various irregular replacements. More than ten PGE minerals were observed to be associated with the sulfide minerals (S.G.U., 1984). The dominant one is sperrylite ($PtAs_2$), which occurs as euhedral crystals (0.005–0.2 mm) at the boundary of pentlandite, chalcopyrite, and pyrite or within chalcopyrite and cubanite. Other PGE minerals include native platinum, moncheite ($PtTe_2$), (Pd,Au), (AuPtPd); palladium melonite (Ni, Pd, Pt)$(Te, Bi)_2$, michenerite (PdBiTe), and sudburyite (Pt, Pd)Sd (S.G.U., 1984).

Sampling and Analytical Procedures

Samples for the present study were collected both from drill cores and underground. Most of the samples which were used for establishing the principal features of the Ni-Cu-PGE mineralization of the deposit were collected from three drill cores. Two drill holes in the west and west-central subchambers (Figs. 2 and 3) intercepted the orebodies from the hanging wall to the footwall, thus providing two transverse sections. Petrological study has shown that the western part of the intrusion probably represents the deeper, more basal part of the magma chamber and could have been the main conduit for the Jinchuan intrusion (Chai and Naldrett, 1992). Study of Ni-Cu-PGE mineralization of samples from along these transverse sections can thus provide constraints on this interpretation. Samples from drill hole zk86 (Fig. 4), which intersected the east subchamber of the intrusion and orebody 2, illustrate the vertical variation of sulfide mineralization.

Nickel, copper, cobalt, and sulfur were analyzed by X-Ray Assay Laboratory, Don Mill, Ontario, Canada, with the X-ray fluorescence method. PGE were determined by the combined fire assay-acid dissolution preconcentration and instrumental neutron activation analysis methods. This method was developed by Hoffman et al. (1978) based on the original method of Robert et al. (1971). Fifty-gram aliquots of powdered sample were mixed with a certain amount of sodium tetraborate and carbonate flux and then heated to 1,000°C for about one hour to produce a PGE-bearing sulfide bead. This bead was then crushed and dissolved in hydrochloric acid. The residue from the dissolution was collected by filter paper and irradiated at the SLOWPOKE neutron reactor. Counting of PGE and Au was performed at the SLOWPOKE Laboratory and the Department of Geology, University of Toronto. All the samples were analyzed in duplicate. Two samples of the standard (SARM) were also irradiated, counted, and calculated as quality assurance standards. Samples with higher PGE contents, roughly more than 50 ppb for Pd and Pt and 5 ppb for Os, Ir, Ru, and Rh, gave results with a less than 20 percent disagreement between duplicates. However, samples with lower values sometimes disagreed by 20 to 50 percent; this is particularly true of Os, Ru, and Rh. A few samples which had a discrepancy between duplicates greater than 50 percent have not been used in this paper.

Analytical Results

The analytical results for Ni, Cu, S, and PGE are listed in Table 1.

Geochemistry of orebody 24

Drill hole zk71 traverses the complete lateral extent of the intrusive rocks and orebody 24 (Fig. 2). The hanging-wall intrusive rocks range in composition from olivine pyroxenite to lherzolite and the footwall rocks from lherzolite to olivine pyroxenite. The orebody is within dunite and lherzolite located at the core of the intrusion. The Ni, Cu, S, and PGE contents, as well as the metal elements in the sulfides along the drill hole, are plotted in Figure 7.

The nickel content is usually below 1 wt percent over this section, and it rises to more than 1 wt percent only at the central part between a 350- to 430-m depth. This is in general agreement with the observation that orebody 24 is dominated by disseminated ore (<1 wt % Ni). The sulfur content fluctuates along the drill hole but is within the range of 1 to 4 wt percent (Fig. 7). Within this section, Ni and Cu correlate positively with S, but their variations are smoother than the latter. The concentration of Os + Ir + Ru also shows a variation trend similar to that of S, except for a very obvious maximum at the central part. The concentrations of Pt, Pd, and Au show two main high value portions; the one between the 350- to 450-m depth has a Pt content of more than 1 ppm. It is interesting to note that the location where sulfur reaches its maximum is also the place for the maximums of Ni, Cu, PGE, and Au, indicating that they are all related to the sulfide content of the ore.

Since sulfides in the ores are overwhelmingly dominated by monosulfides, i.e., the number of metal atoms equals that of sulfur, and also since the atomic weights of the major metal elements (Fe, Ni, Cu, and Co) are similar in a rough approximation, the S content of the sample can be assumed to be linearly

TABLE 1. Analytical Data on the Jinchuan Sulfide Samples

Sample no.	Depth[1] (m)	Location	S	Ni	Co	Cu	Sulfides	Os	Ir	Ru	Rh	Pt	Pd	Au
Orebody 24														
L-16	1,250	Shaft	9.32	2.24	0.07	1.58	25.02	70.2	87.3	76.0	20.5	108.3	194.1	120.3
L-20	1,250	Shaft	4.96	0.99	0.02	2.08	13.59	23.1	24.8	29.3	9.8	446.4	185.4	368.7
L-25	1,250	Shaft	2.07	0.42	0.02	0.83	5.64	12.5	8.2	16.8	15.0	730.5	299.8	513.3
I-13	185	zk71	0.57	0.23	0.02	0.07	1.52	6.5	8.5	9.0	4.7	129.1	73.7	38.5
I-17	270	zk71	3.00	0.56	0.03	0.20	7.85	8.5	6.0	14.1	6.8	659.6	299.5	55.8
I-18	305	zk71	2.23	0.59	0.03	0.30	6.00	6.4	7.4	6.7	3.3	256.7	154.1	114.6
I-20	340	zk71	1.26	0.41	0.02	0.19	3.38	7.5	9.8	11.7	4.7	195.7	93.9	32.7
I-21	360	zk71	6.70	1.73	0.06	0.93	17.97	4.2	3.4	4.0	4.2	55.5	78.7	112.4
I-22	394	zk71	2.60	0.70	0.02	0.36	7.02	14.6	13.6	11.3	8.0	591.5	271.4	192.8
I-23	420	zk71	2.55	0.89	0.02	0.82	7.06	17.5	17.0	20.2	9.5	877.0	302.7	417.3
I-25	465	zk71	1.65	0.49	0.03	0.14	4.39	4.4	4.2	5.8	1.7	129.6	59.7	28.8
I-26	527	zk71	0.26	0.18	0.02	0.01	0.70	1.9	2.1	2.4	0.9	32.0	33.1	1.5
I-27	540	zk71	2.92	0.73	0.03	0.20	7.73	19.7	21.6	27.4	10.6	54.1	115.0	48.8
I-29	580	zk71	2.10	0.62	0.02	0.16	5.63	1.4	0.9	1.2	0.8	127.5	169.3	33.3
Orebody 1														
A2-35	1,250	Shaft	6.94	1.96	0.06	0.15	18.43	62.8	61.3	70.6	35.6	1,514.0	103.0	33.4
2-W-18	1,250	Shaft	6.53	1.56	0.07	0.10	17.47	26.7	19.6	11.4	20.9	3,046.6	510.2	48.7
2-W-6	1,250	Shaft	2.12	0.40	0.02	0.69	5.71	3.9	3.6	7.4	2.1	76.1	105.5	63.6
2W12	1,250	Shaft	1.49	0.38	0.08	0.07	3.94	6.7	6.2	6.7	6.3	65.1	132.0	20.5
S89-1	1,250	Shaft	7.41	2.24	0.07	0.28	19.78	179.4	208.3	224.8	112.9	371.3	126.6	224.5
S89-2	1,250	Shaft	7.08	2.06	0.06	0.84	19.05	71.2	87.8	65.6	32.2	467.8	184.4	56.6
S89-3	1,250	Shaft	6.74	1.98	0.06	1.14	18.24	70.2	63.8	77.3	36.1	128.2	110.8	98.2
S89-4	1,250	Shaft	6.77	1.46	0.05	1.88	18.32	38.1	49.1	51.1	30.0	325.3	271.6	206.5
S89-5	1,250	Shaft	7.40	2.30	0.07	2.71	20.56	37.0	31.7	53.0	26.6	161.5	320.7	96.3
S89-6	1,250	Shaft	6.91	2.03	0.06	0.97	18.64	49.6	41.9	57.8	26.9	134.2	161.6	134.5
S89-7	1,250	Shaft	7.02	2.09	0.07	0.63	18.84	66.4	70.2	82.4	39.0	128.6	199.7	93.0
S89-8	1,250	Shaft	7.72	2.57	0.07	0.41	20.76	70.7	66.5	84.9	62.1	193.0	153.2	94.2
II-89-1	614	zk51	5.34	1.49	0.04	1.03	14.45	19.4	19.9	15.6	13.0	179.1	168.2	103.6
II-89-2	640	zk51	5.36	1.34	0.04	1.50	14.58	12.5	12.8	13.0	8.1	997.4	180.4	162.8
II-89-3	664	zk51	6.62	2.38	0.05	2.01	18.40	14.9	3.6	0.5	0.0	9,217.1	810.0	200.0
II-89-4	700	zk51	7.57	2.33	0.06	1.50	20.62	10.0	7.8	12.8	5.2	303.1	216.9	461.1
II-89-5	750	zk51	6.88	2.14	0.04	0.21	18.37	3.9	6.4	1.8	2.7	1,921.3	319.1	46.7
II-89-6	550	zk51	3.06	0.65	0.03	1.20	8.36	14.7	13.4	16.0	5.5	158.5	95.7	81.3
II-89-7	780	zk51	7.14	1.14	0.03	4.07	19.80	5.2	4.8	0.3	0.0	3,113.9	499.4	474.2
II-89-8	815	zk51	6.15	0.76	0.03	3.80	17.05	7.4	18.9	0.2	0.0	1,985.0	179.0	880.8
II-89-9	857	zk51	5.44	2.05	0.05	2.54	15.44	2.1	10.8	1.0	0.6	3,815.7	671.9	547.1
II-89-11	760	zk51	7.65	3.00	0.06	1.27	21.05	2.5	4.6	0.4	0.0	1,396.9	85.5	142.3
II-89-12	800	zk51	6.95	1.77	0.05	2.04	18.96	8.4	8.8	3.9	6.7	2,212.8	321.0	96.4
II-89-13	870	zk51	7.40	2.53	0.06	1.04	20.13	3.3	5.5	2.2	6.5	500.3	499.1	139.6
II-89-14	920	zk51	10.50	1.84	0.06	0.46	27.46	14.0	4.2	16.7	14.7	514.7	125.3	79.2
II-89-15	975	zk51	8.78	2.92	0.09	0.52	23.63	2.8	4.4	3.1	7.9	251.7	246.0	51.3
II-W-9	490	zk83	10.10	2.99	0.09	2.13	27.51	75.6	77.0	89.6	46.5	147.3	723.6	185.5
II-W-11	565	zk83	6.88	1.88	0.05	0.58	18.37	51.6	60.7	66.5	33.5	87.0	186.1	117.2
II-W-13	610	zk83	7.24	2.13	0.06	0.90	19.50	17.8	26.3	11.8	17.7	135.5	198.4	263.3
II-W-15	650	zk83	7.33	2.00	0.06	1.45	19.84	8.3	12.2	11.1	10.5	107.4	140.9	258.8
II-W-16	670	zk83	2.59	0.54	0.03	0.09	6.77	11.9	12.2	15.0	7.8	54.0	62.7	48.9
II-W-17	685	zk83	7.32	1.74	0.06	2.64	20.08	12.6	6.5	13.5	5.7	3,520.1	565.5	295.3
II-W-19	720	zk83	5.82	1.58	0.05	2.19	16.07	4.6	4.8	10.5	5.4	522.9	230.1	680.9
II-W-21	755	zk83	5.10	1.26	0.05	0.30	13.51	8.4	10.5	11.4	10.3	303.1	164.3	32.4
II-W-22	780	zk83	10.20	1.82	0.06	0.43	26.68	137.4	105.9	161.4	51.5	2,457.8	273.8	140.8
Orebody 2														
A2-32	1,250	Shaft	9.52	2.00	0.07	0.68	25.13	4.7	3.7	5.7	2.4	38.6	19.5	73.1
A2-41	1,250	Shaft	35.60	8.26	0.08	3.22	94.59	1.9	2.6	2.6	11.7	35.6	169.3	28.5
2-E-3	1,250	Shaft	7.44	1.92	0.07	1.46	20.10	3.5	3.2	5.7	2.3	21.7	49.3	30.9
2-E-4	1,250	Shaft	14.50	3.97	0.08	0.83	38.64	12.4	10.2	10.1	9.8	246.3	322.6	30.8
2-E-5	1,250	Shaft	27.00	9.00	0.14	6.36	74.26	14.8	11.8	26.3	33.1	142.3	266.7	197.8
2-E-7	1,250	Shaft	7.65	1.37	0.06	1.83	20.49	2.8	2.5	2.6	1.9	18.3	48.0	41.9
2-E-8	1,250	Shaft	6.21	1.30	0.04	0.34	16.34	7.4	9.7	7.1	5.2	55.1	11.1	10.1

TABLE 1. (*Cont.*)

Sample no.	Depth[1] (m)	Location	S	Ni	Co	Cu	Sulfides	Os	Ir	Ru	Rh	Pt	Pd	Au
						Orebody 2 (*continued*)								
2-E-9	1,250	Shaft	2.24	0.49	0.03	0.06	5.86	3.6	3.0	5.1	13.5	41.0	52.4	21.1
2-E-11	1,250	Shaft	31.20	4.90	0.07	1.07	81.28	3.9	3.6	6.2	4.2	21.4	101.0	29.6
II-89-16	1,250	Shaft	34.20	4.73	0.24	13.50	92.83	144.9	157.2	131.6	52.3	330.4	128.7	36.7
II-7	102	zk86	0.37	0.11	0.02	0.01	0.95	0.5	0.1	2.6	0.0	0.8	37.1	1.8
II-12	237	zk86	0.25	0.17	0.01	0.00	0.66	0.4	0.0	1.6	0.0	0.5	19.0	0.8
II-16	306	zk86	1.96	0.45	0.02	0.55	5.29	0.8	3.2	1.2	1.0	20.4	18.1	21.2
II-17	320	zk86	2.20	0.58	0.02	0.18	5.83	1.9	1.5	2.4	1.3	49.0	35.3	9.9
II-19	355	zk86	7.06	1.70	0.06	0.30	18.66	8.0	8.3	11.6	7.8	106.6	309.3	103.7
II-21	385	zk86	6.48	1.72	0.05	0.71	17.34	1.8	10.0	3.8	5.6	71.0	34.6	54.7
II-22	408	zk86	10.10	2.10	0.07	0.75	26.66	8.3	8.8	13.5	6.4	30.9	98.2	20.4
II-23	425	zk86	10.20	2.82	0.06	0.84	27.26	4.1	5.3	3.5	6.9	145.2	28.4	12.4
II-24	440	zk86	7.23	1.69	0.05	0.83	19.25	2.3	1.8	3.5	2.0	24.7	63.9	20.0
II-26	474	zk86	6.94	1.56	0.03	1.80	18.90	4.5	5.7	6.7	4.1	50.9	32.6	23.4

Values of S, Ni, Cu, CO, and sulfides are in wt percent; PGE in ppb; calculation of sulfide content in whole rock is based on the monosulfide method (Naldrett, 1981)

[1] Samples from locations other than the shaft are from drill holes and the depth is measured from the surface; samples from the shaft are from the 1,250-m level, which is measured from sea level

correlated with that of the sulfide ores. The relative variation of metal elements normalized to the S content of the sample can thus be considered as normalized to the sulfide ores. When these elements are normalized to the S content, PGE, in particular Pt and Au, show an increasing trend from the margins toward the center of the orebody, suggesting a spatial enrichment of PGE in the sulfides. Cu/S ratios vary smoothly, but Ni/S ratios show a slight increase toward the footwall, which may be the result of gravity segregation of sulfide during differentiation of silicate magma.

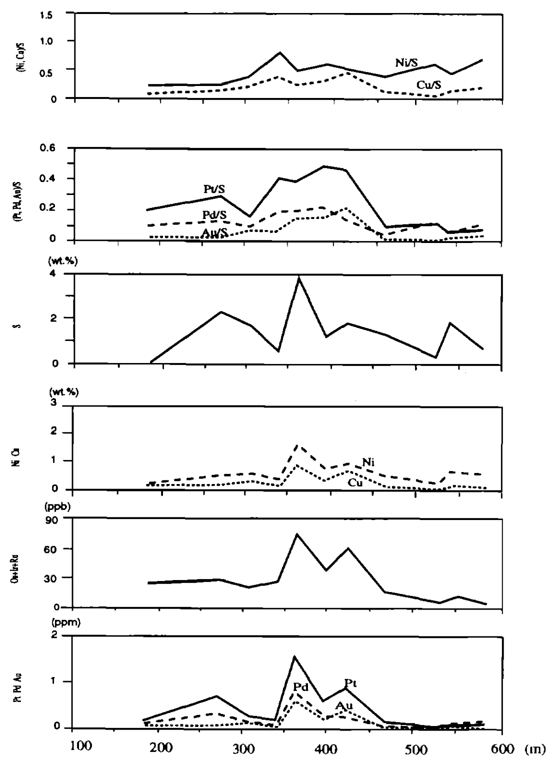


FIG. 7. Metal elements profile through drill hole zk71 which represents a transverse section of orebody 24.

Orebody 1

Drill hole zk51 cuts the major orebody which is comprised almost totally of sulfide-bearing dunite in this section. The variation of the metal elements with depth of the drill hole is shown in Figure 8.

The sulfur content increases relatively steadily with depth, from 3 wt percent at the 550-m depth to 8 to 10 wt percent at the 900- to 1,000-m depth. Nickel is generally between 1 to 2 wt percent in the profile and does not vary much through the whole section except for a slight increase down the drill hole. The asymmetric distribution of the elemental concentration of S and Ni reflects the false transverse section defined by the drill hole. Copper is usually below 1 wt percent except for the portion between 750 and 870 m where it reaches 2 wt percent, and unlike Ni, Cu enriches the central part of the section. Platinum is low in the marginal portions of the orebody, but it forms a plateau between 650 and 870 m. In this portion, the Pt content is usually greater than 1 ppm. Palladium and Au show a distribution similar to Pt, higher at the center and lower toward the margins, but the peaks of Au do not always match those of Pt and Pd. Fluctuations in Pt, Pd, and Au contents in the plateau are also obvious. The other PGE elements

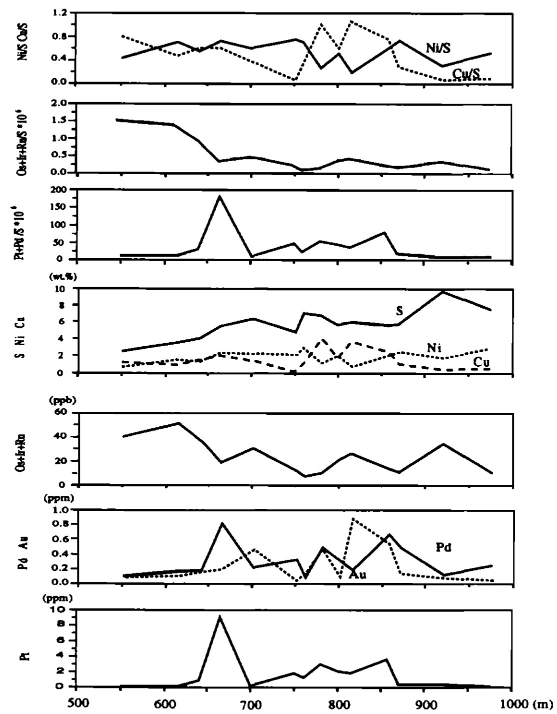


FIG. 8. Metal elements profile through drill hole zk51 which represents a transverse section of orebody 1.

The sulfur concentration is very low in the upper part of the drill hole, corresponding to the unmineralized intrusive rocks, except at about 150 m where there is a small peak of about 3 wt percent S which indicates very weak mineralization. The sulfur content increases from a depth of 320 m down the drill hole to higher than 7 wt percent, which represents the portion of the sulfide orebody. PGE are extremely low in this part with a total content less than 0.4 ppm. This contrasts strongly with their concentrations in the other two orebodies where Pt alone can be greater than 1 ppm. In general, the Pd concentration decreases toward the base of the sulfide zone, the Pt concentration fluctuates irregularly through the sulfide ore zone, and the concentration of Os + Ru + Rh is more or less similar to that of Pt. When normalized to the S content of the samples, Pt + Pd are the lowest at the base of the intrusion and increase gradually upward, reaching a high in the upper part of the net-textured ore and fluctuating in the upper intrusive rocks. The reverse trend of PGE in the sulfide ores to the absolute concentration indicates that the downward increase of the sulfides is more rapid than that of PGE. The Ni + Cu content in sulfide ores increases slightly down the drill hole but is more or less constant throughout the orebodies.

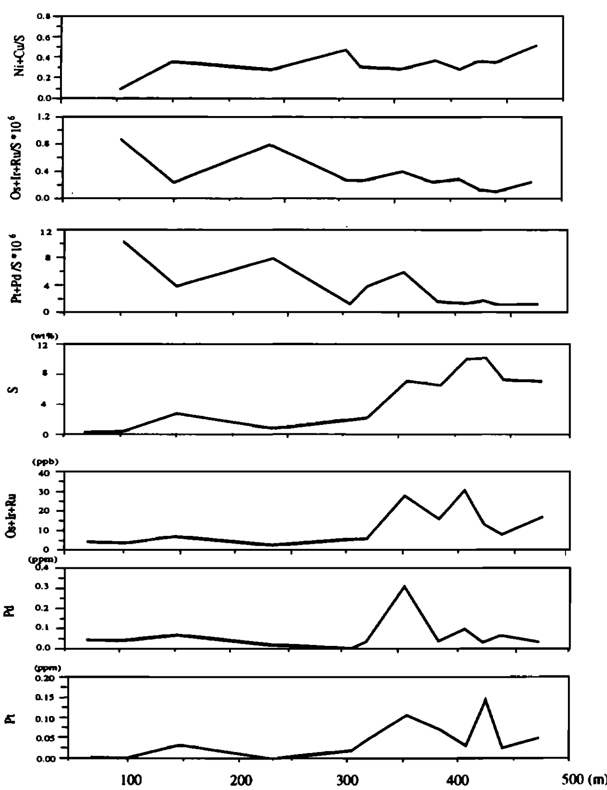


FIG. 9. Metal elements profile through drill hole zk86 which represents a section through the intrusive layering and orebody 2.

such as Ru, Ir, and Os show no regular trends through the profile. Their concentrations are generally one to two orders lower than those of Pd and Pt.

When the metal elements are normalized to the sulfur content of the samples, nickel appears to be flat over the whole profile except for a slightly lower portion between 750 and 850 m. Copper shows a trend of increasing Cu/S from the lower margin to about 750 m, where there is a sharp drop, then the trend increases again until the upper margin. The plot of Pt + Pd demonstrates the same positive central plateau with a large peak at 670 m which is mainly due to the contribution of Pt. The portion with the highest S content does not correspond to the highest Pt and Pd contents or the (Pt + Pd)/S ratio. Unlike other elements, (Os + Ru + Ir)/S ratios reach the highest at the hanging-wall side. This may reflect the contribution of silicate, in particular olivine and chromite, at low S contents.

Orebody 2

Layering occurs in the east subchamber of the intrusion where orebody 2 is located at the base of the igneous body. Drill hole zk86 cuts through the intrusion and intercepts its base and the orebody (Fig. 9). This represents a vertical variation of the intrusive rocks.

Interelemental relationships

All the analyzed samples from different parts of the deposit are considered as a whole to characterize the mineralization of the Jinchuan ores.

Covariation diagrams of Ni, Cu, and Co with S are shown in Figure 10. The amounts of Ni and Co generally increase with S contents, but they do not follow a perfect linear trend in the plot. However, if the sample plots from the western orebodies and the eastern orebody are examined separately, they show two separate sets with a better linear trend but different slopes. Samples from the western orebodies have higher Ni/S and Co/S values than those from the eastern orebody. The linear relationships between Ni and Co and S indicate that their contents in the sulfide ores are more or less constant. The interception of the regression line to the Ni or Co axis is always greater than zero. This is because of the existence of metal in silicates, mainly in olivine since olivine in the sulfide ore contains about 0.2 wt percent Ni (Chai and Naldrett, 1992). The metal content in the sulfide ore should thus be the extrapolation of the linear trends toward that of massive sulfide, i.e., an S content of roughly around 36 wt percent. Based on the data plots, we estimate that the Ni and Co contents in the sulfides are 8.5 and 0.25 wt percent, and 6.5 and 0.15 wt percent, for the western and eastern orebodies, respectively. Cu does not show any clear correlation with S in the Cu-S plot. The highest Cu content can reach about 14 wt percent, but it does not correspond to the highest S content.

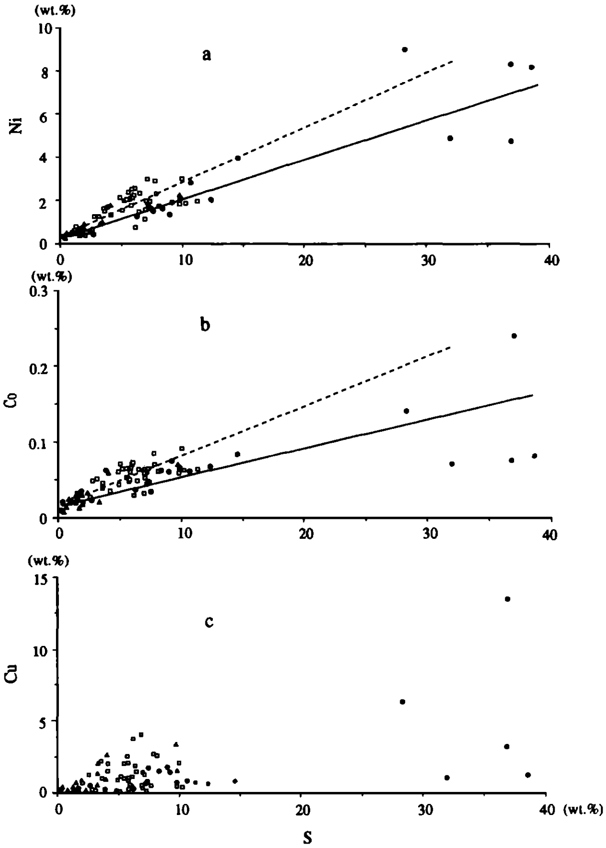


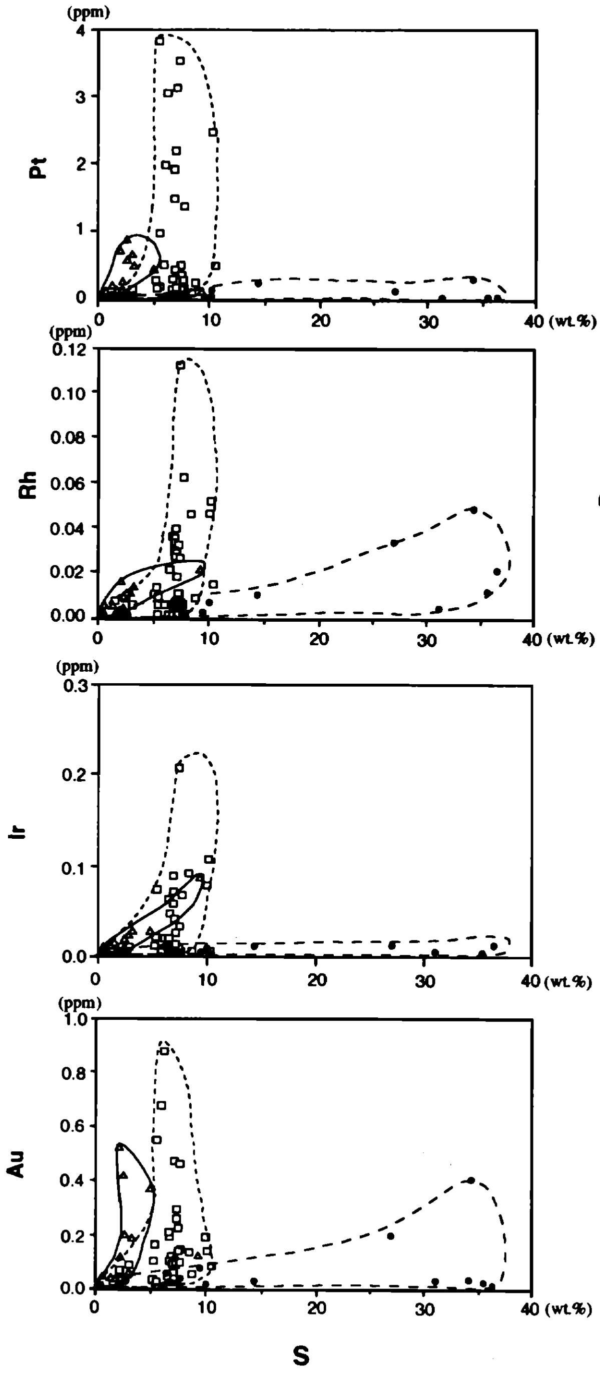
FIG. 10. Correlation of Ni, Co, and Cu with S. Samples from both the western and eastern orebodies show a positive linear correlation between Ni and Co and S, but the Cu plot is generally scattered. Open square = orebody 1, open triangle = orebody 24, and filled dots = orebody 2; Dashed line = regression line for samples from orebodies 1 and 24, solid line = orebody 2.

Overall, massive sulfides have a total PGE + Au content of less than 0.6 ppm. This value is lower than most of the net-textured ore which shows a total PGE and Au content of about 10 to 20 ppm in a 100 percent sulfide fraction. However, the massive ore occurs only in orebody 2 where other ore types are also low in PGE (Table 1). The lower PGE concentration of this orebody may represent PGE concentrated from a magma depleted in PGE. The lower PGE content of the massive ore has also been reported by others who have interpreted the high PGE content in some of the net-textured ore as the result of postmagmatic hydrothermal enrichment (S.G.U., 1984; Yang, 1989).

When PGE and Au contents are plotted against the S content (Fig. 11), the three orebodies show distinct distribution features. Samples from orebody 24 have a low S content. PGE and Au contents increase with the S content, indicating a close association of these elements with sulfide ores. Orebody 1 has a much more scattered PGE plot, even though there is a very vague increase of PGE with S. The scattering of PGE occurs at S contents between 5 and 10 wt percent. At this range of S content, the Pt range is from less than 0.5 ppm to more than 1 ppm, a one order variation in Pt content. Orebody 2 has much lower PGE and Au contents at a corresponding S content than the other two orebodies.

Osmium, iridium, ruthenium, and rhodium, particularly the first three, correlate positively with correlation coefficients (r) greater than 0.95 (Fig. 12). The lower Os, Ir, and Ru tenors of the ores from orebody 2 are also reflected in the diagram in which they are plotted at the lower left corner. It can also be seen from Figure 12 that the Ir/Os and Ir/Ru ratios are 0.97 and 0.85, respectively, which are almost identical to those of the mantle material in which the Ir/Os and Ir/Ru are 1.09 and 0.84, respectively (Morgan et al., 1981; Brügmann et al., 1987).

Naldrett and Duke (1980) constructed a PGE distribution plot in which PGE are recalculated to a 100 wt percent sulfide fraction, normalized to chondrite values, and plotted in order of decreasing melting points. This diagram facilitates the comparison and classification of different types of PGE-bearing deposits (Naldrett, 1981, 1989). PGE data from the



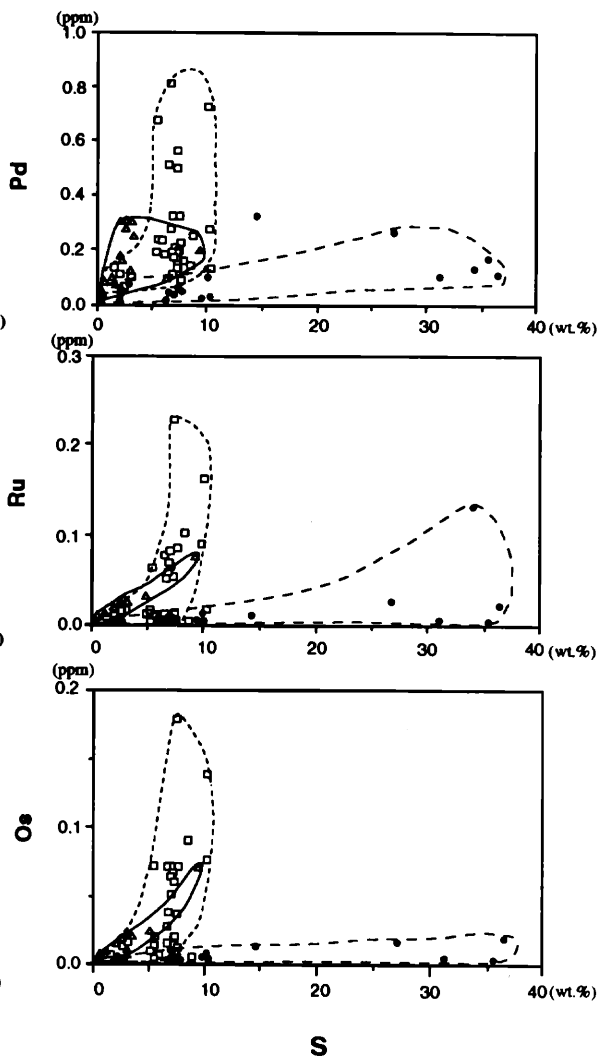


FIG. 11. Correlation of PGE and Au with S. The plot shows two groups: group 1 is high in PGE and Au, representing samples from orebodies 24 and 1; group 2 is low in PGE and Au, representing samples mainly from orebody 2. For legend see Figure 10.

three orebodies of the Jinchuan deposit, together with data of Ni, Cu, and Au, are thus calculated and plotted on a PGE distribution diagram (Fig. 13). The PGE patterns of all the orebodies are very similar, with Pt, Pd, and Au being enriched and Os, Ir and Ru being depleted, whereas Pt shows a slightly positive anomaly for samples from orebody 1. This may correlate with enrichment of Pt at a corresponding S content in this orebody. The difference between samples from the western orebodies and orebody 2 is its lower sulfide-PGE concentration, which is about an order lower than that of the other orebodies.

Discussion

Cause and timing of sulfide segregation

The Jinchuan deposit occurs completely within the Jinchuan intrusion. Sulfide orebodies are present

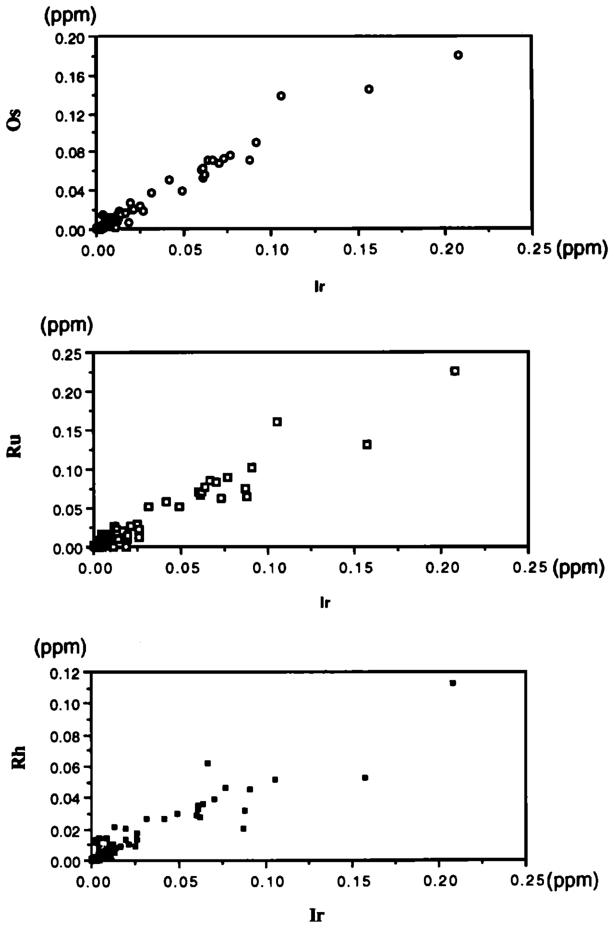


FIG. 12. Correlation of Os, Ru, and Rh with Ir. These elements show good positive linear relationships.

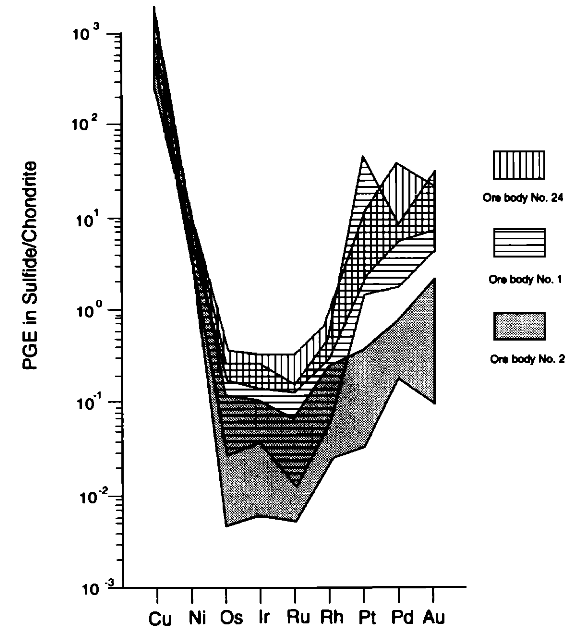


FIG. 13. Chondrite-normalized PGE distribution patterns of ore samples from the three orebodies of the Jinchuan deposit. They are all Pt and Pd enriched and Os, Ir, and Ru depleted. Pt and Pd of orebody 2 are more than one order of magnitude lower than in the other two orebodies, but Ni and Cu concentrations are not so different.

mainly in the lower part of the intrusion and grade from net-textured ore to disseminated ore upward or toward the margins of the intrusion. The Ni content of the ore shows good linear correlation with S. These are typical features of magmatic Ni deposits (Naldrett, 1989).

Sulfur isotope data on sulfide minerals from different ore types of the deposit shown in Table 2 (S.G.U., 1984) indicate that the $\delta^{34}S$ values of the sulfide minerals lie between −2 and +3 per mil. This suggests that the sulfur may have come from the mantle (Naldrett, 1981).

Sulfur to selenium ratios in magmatic Ni sulfide ores have been suggested as an important indicator of nonmantle sulfide contamination (Green, 1978; Barrie and Naldrett, 1989). A primitive magma has a typical S/Se ratio of around 2,700 (Aller, 1967). A ratio of 2,000 to 10,000 is thought normal for magmatic sulfide ores (Naldrett, 1981). The S/Se ratios for the Jinchuan sulfide ores are between 3,000 and 4,000 (S.G.U., 1984) which falls within the range of typical magmatic sulfides.

The Jinchuan intrusive rocks are olivine and chromite cumulates. Variation from sulfide-bearing dunite to other rock types is mainly the result of replacement of interstitial sulfide by pyroxenite or plagioclase. Olivine composition is quite constant with 80 to 85 mole percent forsterite (Chai and Naldrett, 1992). These features indicate that the composition

TABLE 2. Sulfur Isotopic Composition of the Jinchuan Sulfide Minerals

Sample no.	Ore types	Minerals	$\delta^{34}S$ (‰)	$^{32}S/^{34}S$
CBZHS-20	Disseminated	Pyrrhotite	1.8	22.179
CBZHS-7	Net textured	Pyrrhotite	2.2	22.172
CBZHS-4	Net textured	Pentlandite	2.1	22.174
CBZHS-5	Net textured	Pentlandite	−0.3	22.227
CBZHS-5	Net textured	Chalcopyrite	2.5	22.164
CBZHS-6	Net textured	Pentlandite	2.4	22.166
CBZHS-8	Net textured	Pyrite	1.0	22.197
CBZHS-10	Disseminated	Pentlandite	2.5	22.164
CBZHS-10	Disseminated	Chalcopyrite	2.4	22.166
CBZHS-11	Disseminated	Chalcopyrite	2.7	22.160
CBZHS-11	Disseminated	Pyrrhotite	3.1	22.152
CBZHS-14	Massive	Pentlandite	1.6	22.183
CBZHS-15	Massive	Pentlandite	1.8	22.179
CBZHS-16	Disseminated	Chalcopyrite	1.8	22.179
CBZHS-16	Disseminated	Pryite	−2.6	22.278
CBZHS-16	Massive	Pyrrhotite	2.3	22.170
CBZHS-18	Massive	Pyrite	1.8	22.179

Data from S.G.U. (1984)

of the original magma did not change very much during the formation of the intrusive body and that a huge magma chamber should have existed in the intrusion in order to form that amount of olivine with such a narrow range in composition.

Evidence shows that sulfides are always interstitial to cumulus olivine. Sulfides usually occur at the center of the interstitial voids, with interstitial silicates occurring between them and the cumulus olivine. When olivine displays reaction margins in which the olivine crystals are ringed by pyroxene, sulfides commonly occur between the silicate aggregates. With an increasing sulfide content, the amount of interstitial silicate decreases, and in an extreme situation, net-textured ore consists only of cumulus olivine and interstitial sulfide. These features suggest that the sulfide liquid settled later than the cumulus olivine and that pyroxene and plagioclase had not crystallized when the sulfide liquid settled down the cumulus pile. However, it is also not uncommon to see sulfide blebs and inclusions in the cumulus olivine and chromite. These inclusions indicate that some sulfide droplets had already segregated from the magma during crystallization of olivine and chromite. The fact that the composition of the sulfide inclusions is almost the same as the average composition of the sulfide ores indicates that both the sulfide inclusions and the ores are the product of segregation from magmas of similar composition.

From the above discussion, it is clear that the sulfide liquid segregated from a huge amount of magma and that sulfide droplets already existed during crystallization of cumulus olivine and chromite. However, whether sulfide segregation, as well as crystallization of olivine, took place mainly at depth or after the emplacement of magma is still uncertain.

Suppose that batch segregation of sulfide took place at depth and that the silicate magma intruded as an olivine- and sulfide-bearing crystal mush (Jia, 1986). On emplacement of the magma, it is possible that turbulent flow may have broken the sulfide liquid into separate droplets which were kept in suspension. Growth of olivine and chromite might have intensified in response to a change in temperature and/or pressure. Sulfide liquid that became attached to the surface of these crystals could easily have been captured as inclusions. When the magma was emplaced in the present magma chamber, the sulfide liquid, which is much denser than the silicate melt, would have then settled to the base of the magma chamber. The cumulus olivine and chromite could also have been forced down into the sulfide liquid by the weight of the overlying crystals as called upon in the billiard ball model (Naldrett, 1973; Campbell and Naldrett, 1979).

Two observations seem to discount the above interpretation. First, lateral variation of sulfide mineralization is obvious in the western two subchambers in which the sulfide content decreases from the center toward the margins. If the sulfide liquid had settled before the olivine cumulates, penetration of olivine into the preexisting sulfide liquid would have given rise to an evenly distributed net-textured ore at a certain level. Second, the sulfide is enriched in PGE to a different extent at different locations on the same level, which suggests that the sulfide concentrated PGE under different conditions.

An alternative interpretation to early sulfide segregation is that with the intrusion of the silicate magma, sulfide segregated gradually in response to changes in temperature, pressure, and composition due to crystallization of olivine and chromite. A gradual change of the conditions would not produce a sudden segregation of a large amount of sulfide liquid, and the turbulent flow of the magma at the beginning of its emplacement could easily have kept small sulfide droplets in suspension. With further crystallization of the silicate magma, more sulfide liquid would segregate and sulfide droplets would grow by collision of the droplets or addition of further segregated sulfide. To some extent, as a result of less turbulent flow and larger droplet size, sulfides would, as a result of their own density, penetrate into the olivine cumulus pile, driving the interstitial silicate melt upward. The faster cooling rate of the marginal zone magma would increase the crystallization speed of silicates, thus increasing the viscosity of the magma and lessening the penetration of the sulfide liquid toward the margin. This may be the way to interpret the lateral zonation in distribution of the sulfide mineralization. In the upper part, there was a decreasing sulfide segregation due to the exhaustion of sulfide in the magma, as exemplified by the transition from net-textured to disseminated ore and finally to barren rock.

Enrichment of PGE in the sulfides

A high concentration of PGE in the sulfides has been demonstrated in Figures 7 and 8 for orebodies 24 and 1. The figures also show a large variation in PGE concentration across these sections. This cannot easily be explained by batch segregation or deep sulfide segregation as previously proposed (Se, 1985; Yang, 1989). Because of the very high partition coefficients (D) of PGE between sulfide liquid and silicate magma, which is about 10^{5-6} (Campbell and Barnes, 1984), batch segregation or deep segregation would produce high but evenly distributed PGE concentrations in the sulfides. In addition, there would also be a sharp contrast in the PGE concentration between the earlier segregated sulfide and the later in situ progressively segregated sulfide, because the latter was in equilibrium with a PGE-depleted silicate magma. PGE contents in the sulfides are higher at the central part of orebodies 24 and 1 and decrease toward the margins, as shown in Figures 7 and 8, whereas in orebody 2, they increase from the base upward (Fig. 9). These distributions suggest that some process other than batch segregation was responsible for the PGE variations. Fractional segregation of sulfide liquid can give rise to varying PGE contents, depending on the timing and modes of segregation. The earlier segregated sulfide would be in equilibrium with the more primitive magma and would possibly have a higher PGE tenor. On the other hand, the later segregated sulfide would be in equilibrium with PGE-depleted magma and thus contain less PGE. If the sulfide liquid was in equilibrium with different amounts of silicate melt during its segregation or after its settling, a high PGE content could be achieved. The western part of the intrusion appears to have formed by a flow differentiation process in the main magma conduit (Chai and Naldrett, 1992). Olivine, as well as sulfide liquid, would tend to concentrate at the central part of the conduit. A fresh injection of magma would exchange with the interstitial silicate melt and would make contact with the sulfide liquid already settled. Thus, since sulfide in the central part of the conduit would have had more chance to come into contact with the fresh magma, it would have had a greater chance to incorporate more PGE from the passing magma and increase its PGE content. Sulfide away from the center would have had less contact with fresh silicate magma and would thus contain a lesser amount of PGE.

Contrast in PGE concentration in orebody 2 and the other two orebodies

One distinguishing feature of the Jinchuan ores is that the average PGE concentration of the sulfides from orebody 2 in the eastern part is about an order of magnitude lower than those from orebodies 1 and 24 (Fig. 13). If we ignore the irregularities caused by the possible redistribution of PGE, the above observation implies that different magmas or processes must have been involved. The close association of the PGE with the sulfide ores in general and the fact that almost all the PGE-enriched parts (PGE > 1 ppm) occur in the net-textured ore suggest that PGE can generally be considered to be incorporated by the sulfide liquid during segregation. Moreover, analysis by in situ acceleration mass spectrometry on the sulfide minerals has revealed that large quantities of Pd, Ru, Rh, and Os are distributed in pentlandite and pyrrhotite (Chai et al., 1992) and indicates that magmatic sulfides scavenged PGE. As discussed above, concentration of PGE in sulfide is dependent on the availability of PGE in the reservoir (the amount of PGE in original magma from which the sulfide melt segregated) and the volume of magma with which the sulfide melt has equilibrated. Thus, two explanations are possible for the difference of the PGE tenors in different parts of the Jinchuan deposit: the sulfide liquid equilibrated with two batches of magma with constant composition but at different R factors, and the sulfide liquid equilibrated with variably evolved and variably PGE-depleted magma.

The relationship of the concentration of a certain element, i, in sulfide liquid segregating as one batch from a silicate magma can be related to the original (o) concentration of the i element in the magma (X_{i_o}) by the expression (Campbell and Naldrett, 1979):

$$Y_i = D_i X_{i_o} (R + 1)/(R + D_i), \qquad (1)$$

where D_i is the partition coefficient of the i element between a sulfide and a silicate liquid, Y_i is the concentration of the i element in sulfide, and R is the ratio of the silicate melt over the sulfide melt with which the silicate melt was in equilibrium.

Since $R \gg 1$ in the natural system, the above equation can be written as:

$$Y_i = D_i X_{i_o} R/(R + D_i). \qquad (2)$$

Suppose that the Jinchuan sulfide ores originated from silicate magmas with similar PGE concentrations and that the different PGE contents in the sulfides of different orebodies are due to different R factors. The magnitude of these R factors can be estimated according to equation (2). For simplicity, only the Pt content of orebodies 1 and 2 is used here as an example for the calculation. From Figure 11, the Pt contents in sulfide which contains about 36 wt percent S can be estimated as 20 and 0.5 ppm for ore bodies 1 and 2, respectively. Petrological study of the Jinchuan intrusion indicates that the primary magma for the Jinchuan intrusion is a high Mg (12 wt % MgO) basaltic magma (Chai and Naldrett, 1992) which is comparable to the magma of the Bushveld Complex. This complex contains about 15 ppb Pt in its original magma (Davis and Tredoux, 1985). If we assume that the Jinchuan primary magma also contains about 15 ppb Pt and that the distribution coefficient between the sulfide and silicate magma is $D_{Pt} = 10^5$, then the R values can be calculated using equation (2). These are 30 and 1,333 for orebodies 2 and 1, respectively. We admit that the exact results depend heavily on X_{Pt_o} and D_{Pt}. However, the difference in R values for orebodies 1 and 2 will always be very large if we assume that the primary magmas responsible for sulfide orebodies have similar PGE contents.

On the other hand, since PGE are closely related to Ni sulfide, the above-calculated R values also have to explain the Ni contents in sulfides which, as shown in Figure 10, do not vary greatly among all the orebodies. Based on the above-calculated R value, the Ni contents in sulfides can be calculated as about 8.0 and 1.0 wt percent Ni in sulfide for orebodies 1 and

2, respectively, if D_{Ni} and X_{Ni_0} for Ni are assumed to be 275 and 350 ppm, respectively (Naldrett, 1981). The calculated Ni content of 8.0 wt percent for orebody 1 is only slightly lower than the analytical results for the sulfide ores, but a value of 0.95 wt percent Ni in the sulfides for orebody 2 is far too low. Therefore the Jinchuan sulfides cannot all have originated from magmas with a similar PGE concentration, given the above assumptions.

In the previous section, we have estimated that the Ni content in sulfides of orebody 1 is about 8.5 wt percent, and that of orebody 2 is about 6.5 wt percent. These values can be used to put constraints on the calculation of the corresponding R values for the sulfide ores by applying equation (2). Calculation shows that R values of 2,078 and 572 are needed to form the sulfide ores in orebodies 1 and 2, respectively. Similarly, these R values should be suitable for the calculation of the Pt contents of the original magmas. Based on equation (2), Pt concentrations of 9.6 and 0.9 ppb are obtained for the primary magmas which are responsible for orebodies 1 and 2.

Although the above result only represents a rough estimation, it can explain both the observed Ni and PGE features of the sulfide ores. Therefore, it is reasonable to suggest that the orebodies originated mainly from two magmas with different primary PGE concentrations. A Pt concentration of around 9.6 ppb is probably the original concentration of the Jinchuan magma which is not much different from that of 15 ppb for the Bushveld parental magma and the estimated 12 ppb for the Stillwater parental magma (Crocket, 1979). It is thus typical of primitive mantle-derived magma. The Pt concentration of around 1 ppb for magma that gave rise to orebody 2 is much lower than the value of primitive basaltic magma. Our conclusion is that the sulfides of the west orebodies, 24 and 1, separated from a primitive magma and that those of the east orebody, 2, derived from a PGE-depleted magma.

The nature of the primary Jinchuan magma as determined from PGE data

Since the Jinchuan igneous body consists of only ultramafic rocks, the Jinchuan deposit was previously interpreted to be an ultramafic-related deposit with a parental magma having greater than 30 percent MgO (S.G.U., 1984; Jia, 1986). The study of the petrology and geochemistry of the Jinchuan deposit by Chai and Naldrett (1992) has, however, posed questions concerning the ultramafic affinity of the Jinchuan deposit. The typical cumulate texture, the moderate forsterite content (80–86 mole %) of olivine, and the light rare earth element enrichment of the intrusive rocks have led to the suggestion that the Jinchuan igneous body was originally the cumulate part of a gabbroic intrusion.

Results of the present study provide further support for this conclusion. Experience has shown that there is a progressive increase in the (Pt + Pd)/(Os + Ir + Ru) ratio from ultramafic to mafic magmas and thus in related sulfide deposits (Naldrett, 1981). Naldrett (1981) has indicated that sulfides associated with peridotitic komatiites, which have crystallized from magmas containing more than 20 percent MgO, have a (Pt + Pd)/(Os + Ir + Ru) less than 2.0, and that ratios greater than 13 are characteristic of magmatic sulfide deposits related to gabbroic rocks that have crystallized from magmas containing less than about 12 percent MgO. (Pt + Pd)/(Os + Ir + Ru) values of the Jinchuan sulfide ores fall between 10 and 15. These are close to the typical ratios of sulfide ores derived from gabbroic magma.

Barnes et al. (1988) have devised a Pd/Ir and Ni/Cu diagram which illustrates the effect of different partial melting of mantle material on the distribution of PGE, Ni, and Cu. Since Pd/Ir increases as magma suites become more evolved, while Ni/Cu decreases (Fig. 14), a plot of these two ratios against each other can produce a separation of the magma suites, with the most primitive (mantle material) at one end and the most evolved (continental flood basalts) at the other. The effect of variations in the degree of partial melting and the variation in composition of the source material on the Pd/Ir and Ni/Cu ratios can be estimated empirically from the figure.

In a plot of Ni/Cu (0.2–10) and Pd/Ir (1–100) ratios, all the Jinchuan ore samples fall into the area of high Mg basalts and flood basalts with the majority falling in the high MgO basalts region (Fig. 14). Although the olivine cumulate usually contains a certain amount of Ni, which would displace the data more to the right closer to the ultramafic area, the data still plot in a distinctly different field from that of ultramafic rocks. This supports the results of the pet-

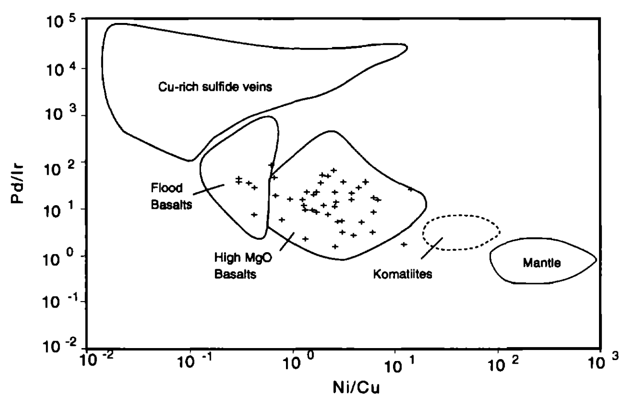


FIG. 14. Plot of Pd/Ir against Ni/Cu of the Jinchuan sulfide ores (+ symbols). The outlined areas are defined by Barnes et al. (1989).

rologic study in which the Jinchuan intrusion was concluded to have originated from a high MgO basaltic magma (Chai and Naldrett, 1992).

Chondrite-normalized PGE distribution patterns can be used to characterize PGE mineralization of different types of PGE-bearing sulfide deposits (Naldrett, 1981; Naldrett and Barnes, 1986). PGE in ultramafic-related Ni deposits usually have low, flat PGE distribution patterns whereas PGE in mafic-related deposits commonly show Pt, Pd, and Au enrichment that to some extent reflects the PGE contents in the original magma from which the sulfide melt segregated and with which it equilibrated (Naldrett and Von Gruenevaldt, 1989). The PGE patterns of all three orebodies of the Jinchuan deposit are similar, with Pt and Pd enrichment and Os, Ir, and Ru depletion, except for the lower concentrations in orebody 2 (Fig. 13). When compared to other typical gabbroic PGE and PGE-bearing sulfide deposits in the world, the PGE distribution pattern of the Jinchuan ores shows features very similar to these deposits and overlaps the field of the Sudbury deposits (Fig. 15).

Role of hydrothermal activity

PGE-bearing deposits are generally considered to be the result of the segregation of sulfide liquid from a mafic or ultramafic magma (Naldrett, 1981; Keays et al., 1982; Naldrett and Barnes, 1986). However,

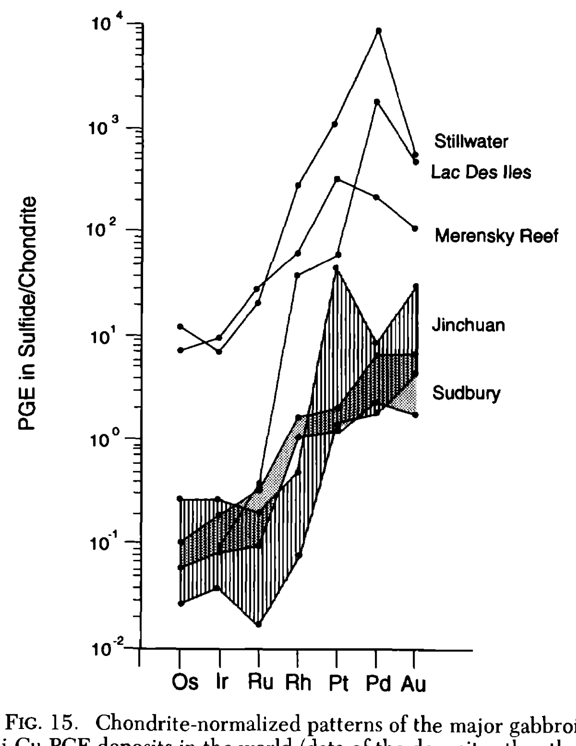


FIG. 15. Chondrite-normalized patterns of the major gabbroic Ni-Cu-PGE deposits in the world (data of the deposits other than Jinchuan are from Naldrett, 1981).

questions on the role of late-stage magmatic or hydrothermal fluids in the concentration of PGE have also been raised (Stumpfl and Rucklidge, 1982; Boudreau and McCallum, 1985; Ballhaus and Stumpfl, 1985; Rowell and Edgar, 1985).

The Jinchuan deposit has shown characteristic features of Ni-, Cu- and PGE-bearing sulfides. PGE mineralization is always associated with sulfide ores. Whole-rock PGE concentrations increase with increasing S, i.e., with the increasing sulfide contents of the samples. The Pt and Pd enrichment is always associated with the enrichment in Os, Ir, and Ru, which commonly resist hydrothermal remobilization. PGE enrichment is absent in orebody 2 where the PGE content in the sulfides is very low in comparison with the other orebodies, although hydrothermal alteration and fracturing of the rocks is similar to that in the western part of the intrusion. Moreover, pentlandite is reported to have contained high concentrations of Pd, Os, Ru, and Rh (Chai et al., 1992). These features strongly suggest a general magmatic origin of PGE mineralization.

However, we have also observed that the breccia zones in the sulfide orebodies can be enriched in PGE, particularly in the western part of the deposit, and that most of the Pt occurs as sperrylite ($PtAs_2$). These features indicate that hydrothermal fluid did play a role in PGE mineralization (Sun, 1986; Yang, 1989). Another observation from this study is that PGE can vary from a few ppm to less than half of a ppm (Fig. 11). This variation in PGE concentration may partly be accounted for by the variation of the *R* factor at different locations of the magma chamber as discussed in the previous section. Sulfides in the central part of the magma conduit can achieve higher *R* values and would thus contain higher amounts of PGE, but a quantitative explanation is difficult based on the *R* factor. Therefore, PGE, in particular Pt, may have been redistributed by postmagmatic hydrothermal fluids. However, these enriched or depleted spots are still in the sulfide ore and if hydrothermal activity played a role in concentrating PGE, the PGE must have been derived from the sulfide ore that was magmatic originally. Studies by acceleration mass spectrometry on the dominant Jinchuan sulfide minerals indicate that pentlandite can account for most of the Pd, part of the Os, Ru, and Rh, and that almost all of the Pt, Ir, and Au form independent minerals, possibly metal sulfides and alloys (Chai et al., 1992). Since these minerals are very fine grained and are located at the boundary of the main sulfide minerals, hydrothermal fluids that contain As, Sb, and Te, which have a high affinity for PGE, may have reacted with the PGE alloys and sulfides to form arsenides, tellurides, and stibnites during hydrothermal activity. The PGE enrichment along breccia zones may indicate that some PGE move with the fluid. However, since only the breccia zones in the sulfide ore can be enriched in PGE, this appears to indicate that PGE cannot be moved too far away from the original site of precipitation. Therefore, hydrothermal fluids are not the original agent for the concentration of PGE in the deposit.

Copper does not show a good correlation with S (Fig. 10). Chalcopyrite in the ore commonly crosscuts silicate and other sulfide minerals. This feature may also be due to remobilization of Cu by hydrothermal fluids.

To summarize the above discussion, hydrothermal activity took place in the Jinchuan deposit. It affected the distribution of Cu and PGE but did not play a major role in concentrating PGE in the Jinchuan deposit. It may have caused very local redistribution of PGE, Au, and Cu and may have formed different PGE minerals.

Genetic model for the PGE mineralization of the Jinchuan deposit

A hypothetical model consistent with the geology and geochemistry, in particular, the PGE mineralization, of the Jinchuan deposit is constructed and shown schematically in Figure 16.

Tectonic rifting resulted in the intrusion of a high Mg basaltic magma in the Jinchuan area. This magma may have had olivine and chromite crystals, as well as small amounts of sulfide droplets, in suspension during its intrusion (Chai and Naldrett, 1992). As it passed through the main conduit, which is the western part of the intrusion, olivine and chromite settled out because of their higher density. However, a narrow conduit and rapid velocity of magma can keep the suspended crystals from settling down directly to the base of the magma chamber. In this case, magma differentiation is controlled by flow differentiation that concentrates the cumulus minerals in the central part of the conduit (Barker, 1983). In the same way, sulfide droplets also tend to concentrate at the center, but the turbulent flow of the magma prevents them from growing to any appreciable size.

At Jinchuan above the magma conduit, the upward current of the magma decreased quickly because the magma chamber was wider, allowing the magma to spread out laterally. As a result of this, olivine and chromite settled to join a growing pile of cumulates and sulfide droplets also settled with the olivine and chromite. Fresh injected magma tended to exchange with evolved interstitial magma, cumulus olivine and chromite crystals overgrew, and further segregated sulfide liquid was added to the preexisting sulfide liquid, thus increasing the amount of interstitial sulfide at the central part of the conduit. On the other hand, as the fresh magma flowed through the cumulus pile, it was also brought into contact with the interstitial sulfide liquid, thus increasing the R value of the sulfide. As more silicate magma passed by the center of the conduit, sulfide in this part could achieve a larger R value and thus contain a larger amount of PGE. Toward the margins, the magma flow was slower in velocity and less in quantity. The sulfide liquid in these parts had less chance to come into contact with the silicate magma and would contain a smaller amount of PGE. Therefore, a lateral zonation of PGE in sulfide was developed (Figs. 7 and 8).

In the east subchamber, which is the location of orebody 2, the appearance of massive sulfide indicates that there must have been an abrupt change of conditions in the magma chamber. The extremely low PGE concentration in this massive sulfide precluded the possibility of sulfide segregation from primitive magma. Chai and Naldrett (1992) concluded that the western part of the Jinchuan intrusion represents the main conduit of the magma chamber and the eastern part represents the higher level of the intrusion. Based on the previous discussion, the magma that reached the upper level was depleted in PGE as a result of equilibration with the sulfide liquid at the conduit. The change from massive, to net-textured, to disseminated ores upward suggests that conditions for sulfide segregation gradually became normal. In order to achieve a sudden saturation of sulfide, there should be either an addition of S from outside the system such as in the Duluth Complex, (Ripley, 1981), contamination by more felsic country rocks as in Sudbury, Canada (Naldrett, 1989), or a mixing of magmas (Campbell and Turner, 1987). As shown previously, neither S isotope data nor S/Se ratios support the addition of S from an outside source. There is no evidence of serious country-rock contamination. The vertical phase layering of the rock types in the east subchamber does imply periodic replenishment of silicate magma (Chai and Naldrett, 1992). Therefore, it is very likely that formation of massive sulfide in orebody 2 was due to magma mixing. At some stage, a new pulse of magma, depleted in PGE, down to the 1 ppb level as calculated, intruded into the east subchamber and encountered the evolved magma. Since the new silicate magma was more primitive in terms of major element composition and thus was heavier, it would likely be injected as a magma fountain and spread above the olivine cumulate pile (Campbell and Tunner, 1987). The mixing of the two magmas would cause an oversaturation of sulfide and the sulfide liquid would likely settle to the bottom of the magma chamber. The R factor of this sulfide liquid was smaller at the start due to the large amount of segregation, and thus the PGE concentration in the sulfide liquid would be lower. As magma mixing continued, segregation of sulfide liquid would decrease. Later segregated sul-

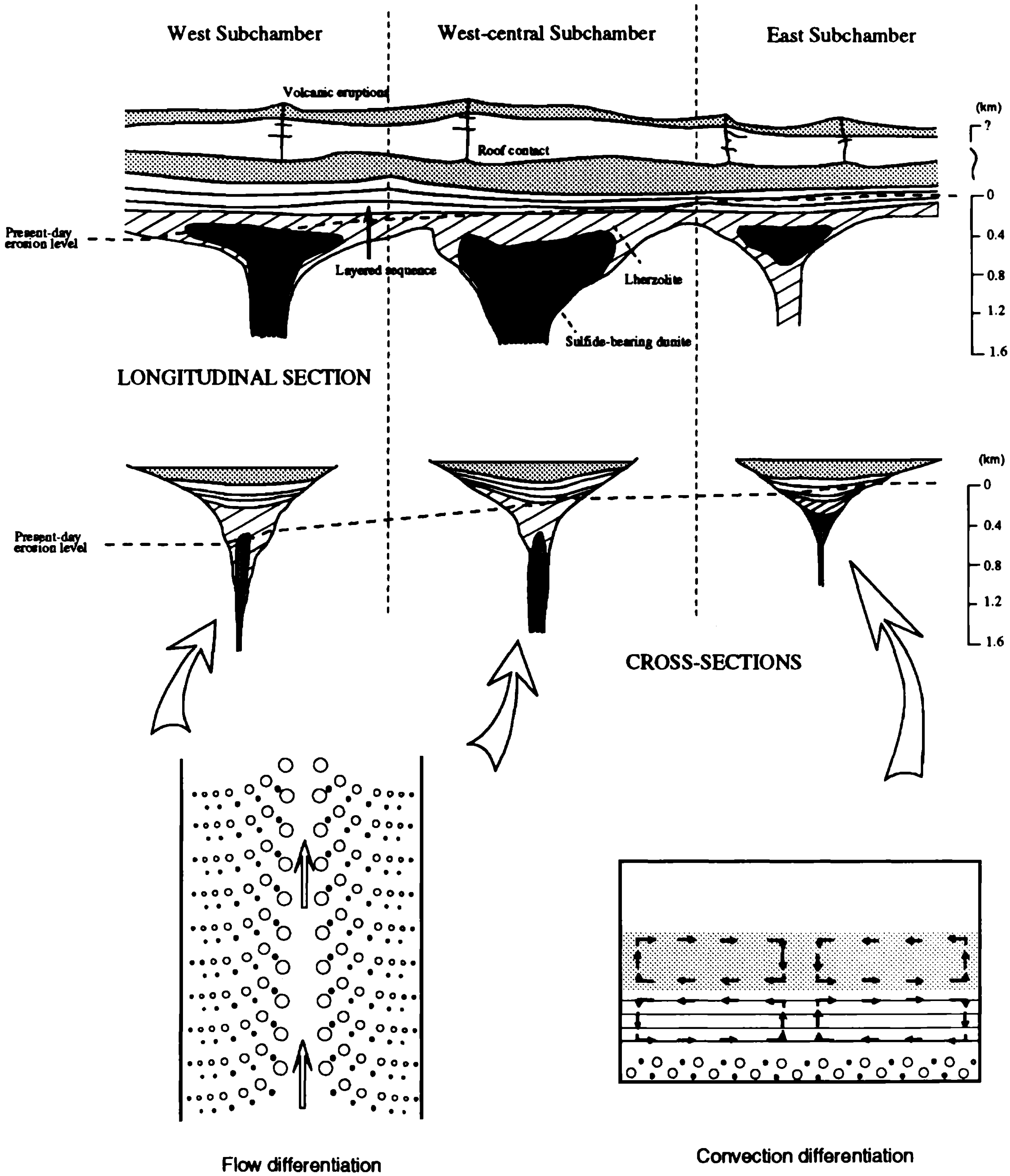


FIG. 16. Genetic model for the PGE mineralization of the Jinchuan deposit. The upper diagram represents an idealized longitudinal section and the lower two demonstrate the differentiation of silicate magma and segregation of sulfide liquid. Open circle = cumulus olivine, solid dots = sulfide droplets.

fide could achieve higher R values, but this was counterbalanced in part, but not entirely, by the effect of the depletion of PGE in the silicate magma. Therefore, even though the absolute amount of PGE decreased upward due to the rapid decrease in sulfide concentration, PGE contents in 100 percent sulfide increased as shown in Figure 9. The surge of sulfide and PGE contents in the upper part of the intrusion may indicate injection of another relatively smaller pulse of magma.

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